DATA MODELLING & FORM DESIGN

Practical Implementation Report — Hotel Booking Management System

DOMAIN	PLATFORM	ARCHITECTURE
Hotel Operations	Low-Code / Zoho Creator	Relational (3NF)

1. Aim

To perform relational data modelling up to Third Normal Form (3NF) for a Hotel Booking Management System and build responsive user interface forms and operational analytics dashboards for efficient data entry, management, and report generation.

2. Procedure

1. **Requirements Analysis & Entity Identification:** Defined core domain entities including `Booking Details`, `Payment Details`, `Guest Details`, `Room Details`, and `Check In/Out Records` along with their key attributes.
2. **Schema Normalization (3NF):** Applied normal forms to eliminate transitive dependencies and redundant fields, enforcing data integrity via Primary Keys and Foreign Key relational lookups.
3. **UI Form & Field Mapping:** Constructed data entry forms (e.g., *Booking Details Form*) utilizing appropriate control types (Lookup Selects, Date Pickers, Currency inputs, and Numeric fields).
4. **Dashboard & Metrics Configuration:** Built visual metric widgets ($1,840.50 Total Amount, Booking Source counters) and dynamic analytical graphs to display real-time operational insights.

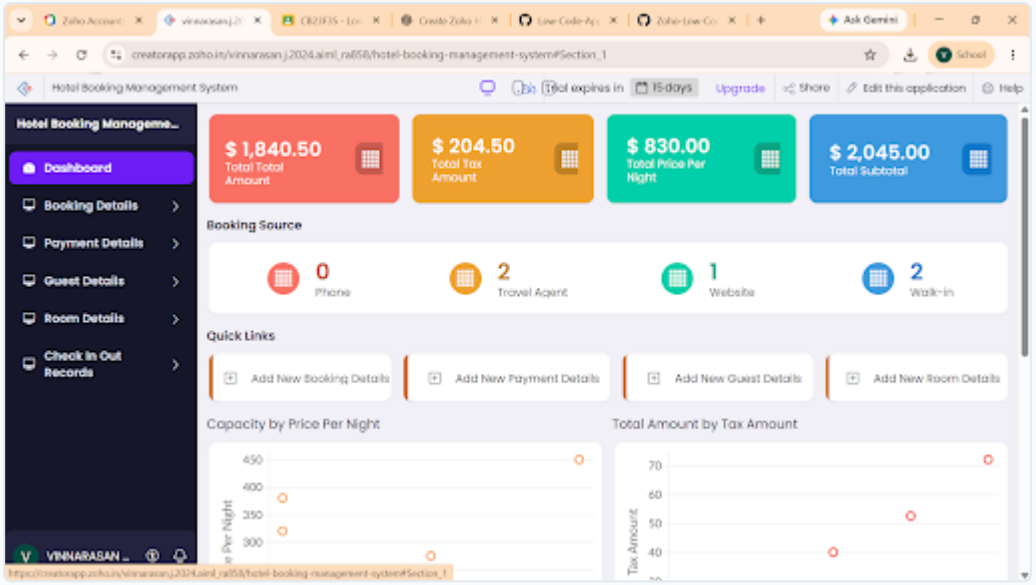


Figure 1: Main Application Dashboard displaying financial aggregates, booking channel counts, and analytical charts.

3. Application Forms & Data Reports

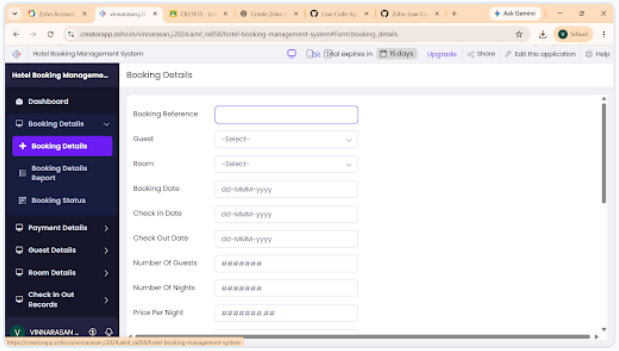


Figure 2: Data Entry Form for Booking Details showing lookup pickers and input controls.

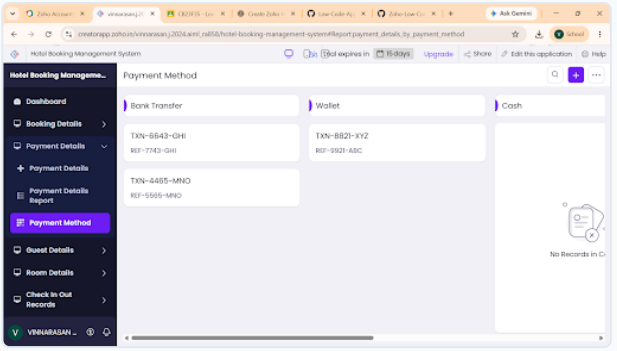


Figure 3: Categorized view of payment records grouped by method (Bank Transfer, Wallet, Cash).

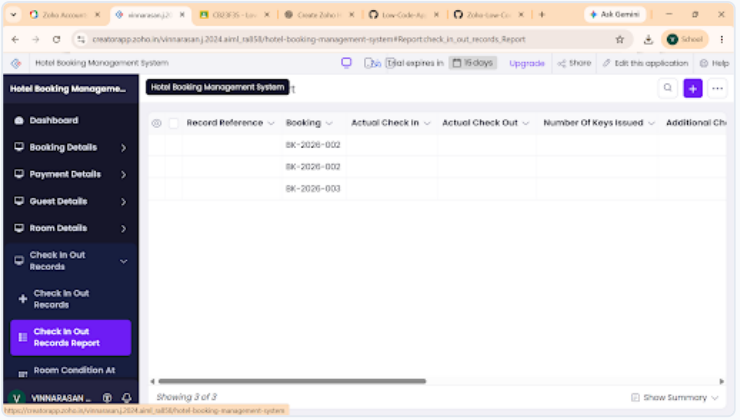


Figure 4: Tabular report interface for Check-In / Check-Out verification and key tracking.

4. Result

The **Hotel Booking Management System** data model was successfully created and normalized to 3NF. Interactive user interface forms were built and integrated with back-end database entities. Key outcomes include:

Normalized Database Schema

Eliminated data redundancy and ensured operational integrity through enforced foreign key relationships across bookings, guests, and payment methods.

Functional UI & Analytics

Delivered smooth data entry forms with real-time validation alongside visual dashboard reporting for operational visibility.